

How Workflow-Driven Automation Becomes Operable

under Air Gap and Compliance Constraints

AUTOCON S

THE NETWORK AUTOMATION CONFERENCE

Who we are?



Christian Miene
Network Automation Engineer



Henry Ölsner
Principal Technical Consultant



The first meeting...

The first meeting...

“We want to automate our **brownfield environment** within our **critical infrastructure**”

“We have a lot of **existing python scripts** on our **backup server** that we need to **operate**”

“We developed a **reporting web-app** that needs to be deployed”

Let's deploy it
quickly on Friday

Let's deploy it quickly on Friday

No connectivity to GitLab

The python application requirements vs. legacy enterprise repository versions

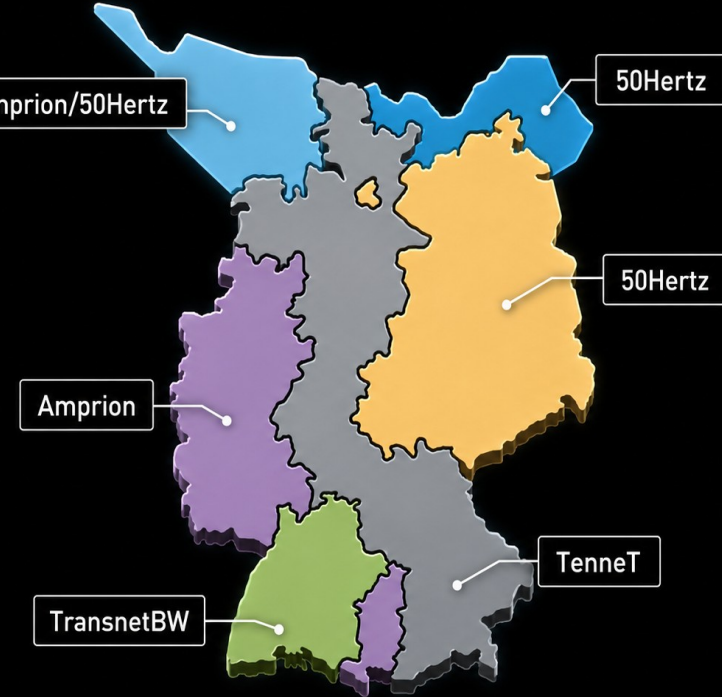
air-gapped installations → Wheels → transitive dependency chain

Insufficient permissions to resolve dependencies

What is the
environment?

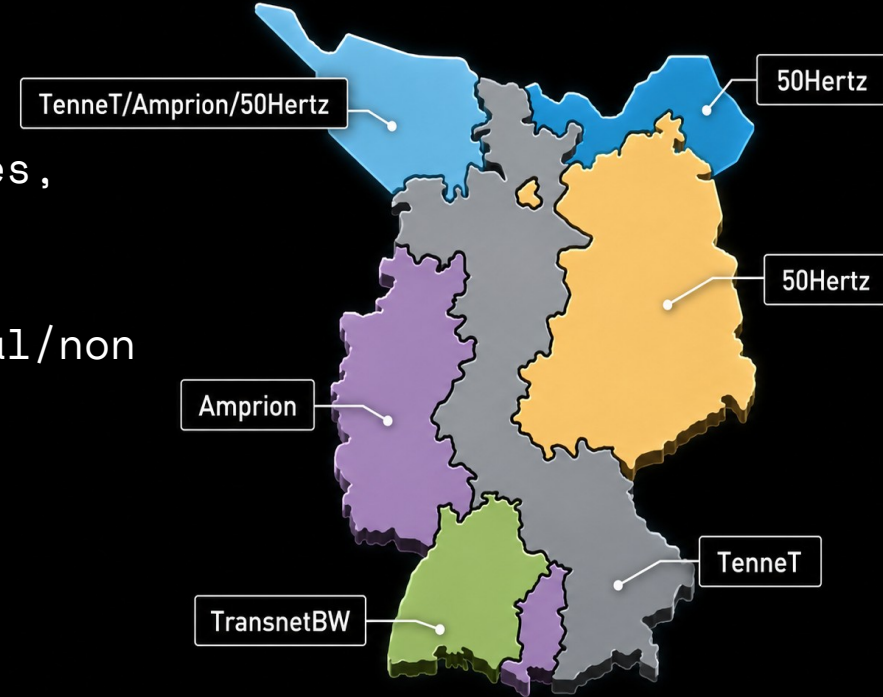
50Hertz Transmission GmbH

- Transmission System Operator (TSO)
- Operates critical high-voltage energy infrastructure
- Part of Germany's Critical Infrastructure

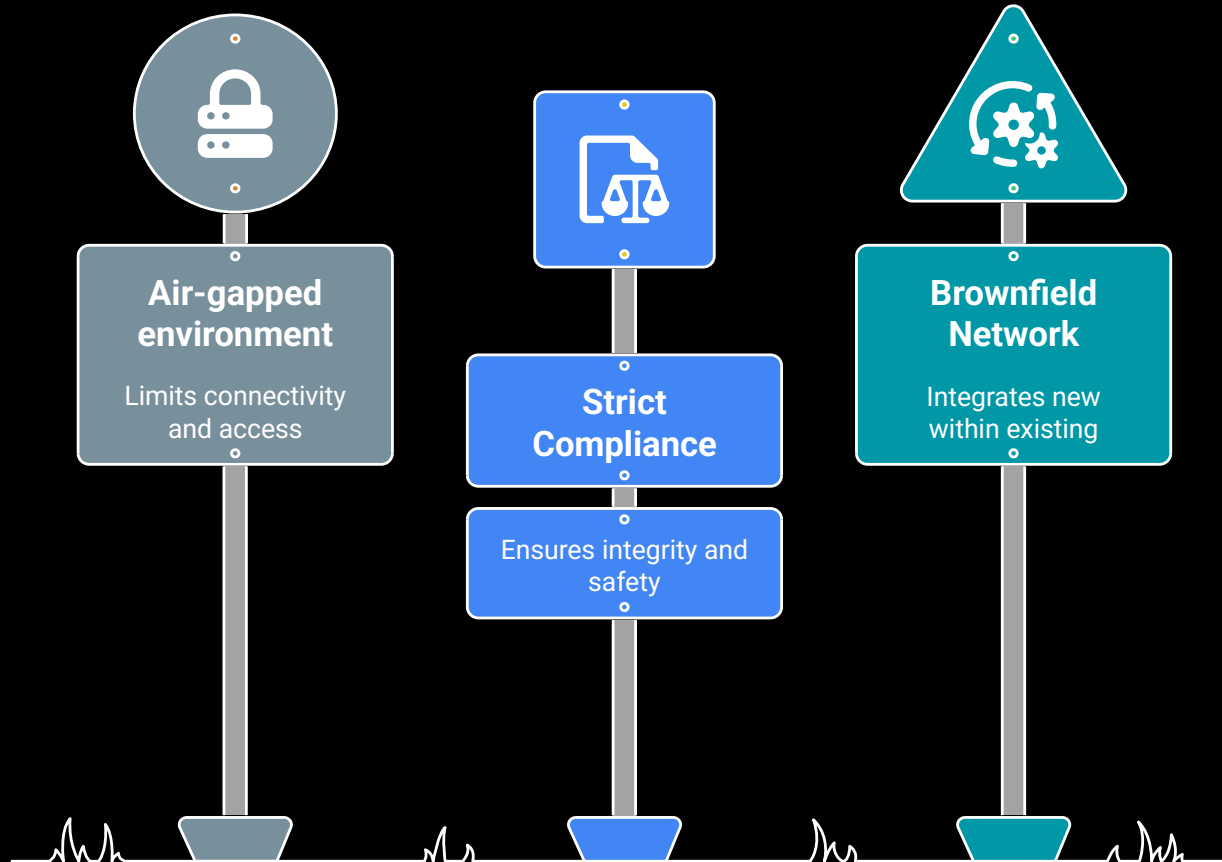


50Hertz Transmission GmbH

- around 4,000 IT infrastructure devices: routers, switches, firewalls, load balancers, proxies, DDI, etc.
- 6 Network Zones/Domains (critical/non critical)



The Operational Constraints



Where we are today?

5

|

4

|

719

Team Members

Apps

Workflow Executions

106,654

Sub-process executions

As of 29.05.2026

Implemented Use Cases until Today

VM Network Rollout

Setting up VM networks with end-to-end connectivity and documentation

CMDB Synchronization

Aligning CMDB with management and monitoring tools

802.1X Migration

Automating the migration to 802.1X authentication

MPLS VRF Rollout

Rollout of MPLS VRFs for WAN network segmentation

Data Center Self-Service Portal

self-service portal for switch port configuration and checks



How did we get there?

Where our automation journey started



What We Had

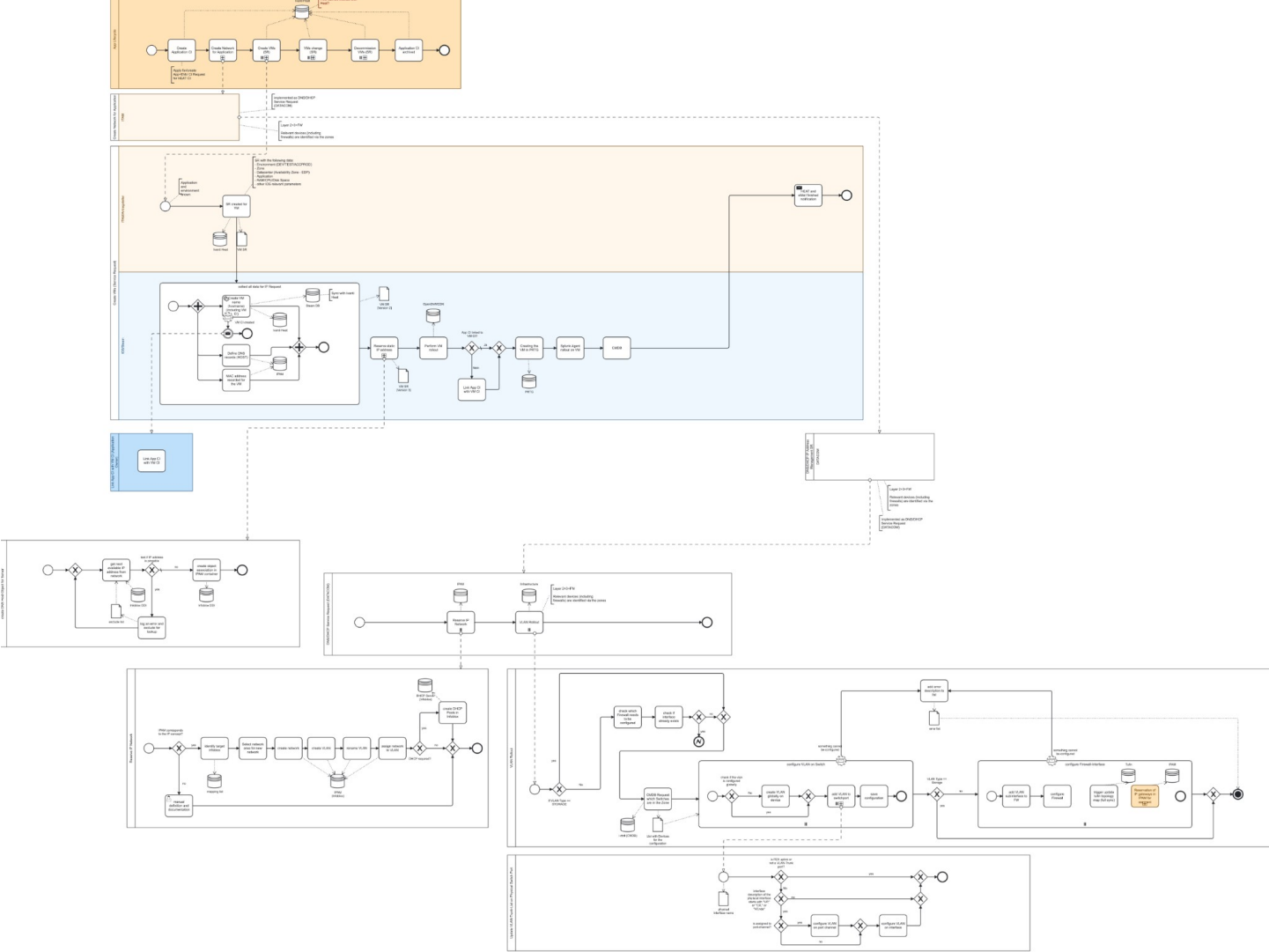
python scripts like test_final.py
Large unmanaged log files
Manual SSH sessions
Regex-based parsing
Undocumented cron jobs

Use-case inventory in Excel
Classification and grouping
Prioritization by business value
First stakeholder: server team

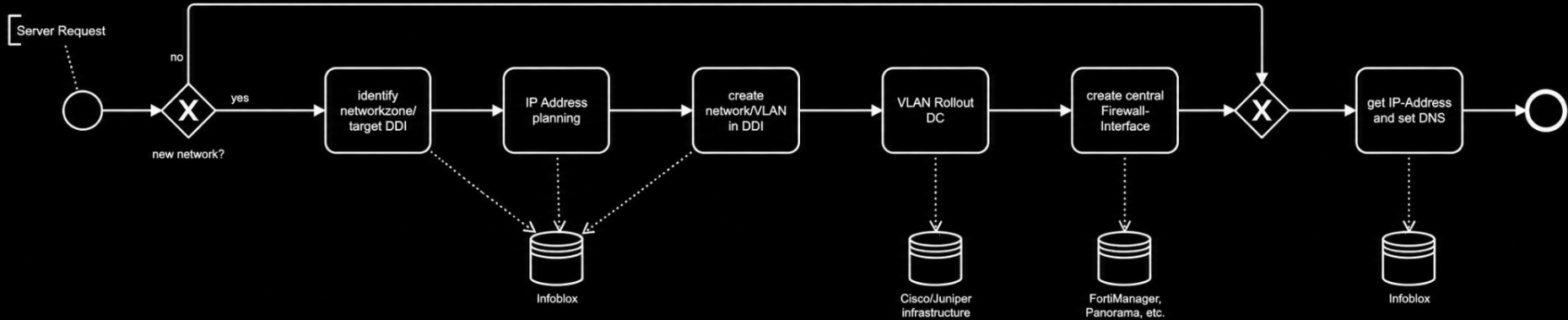
How We Started



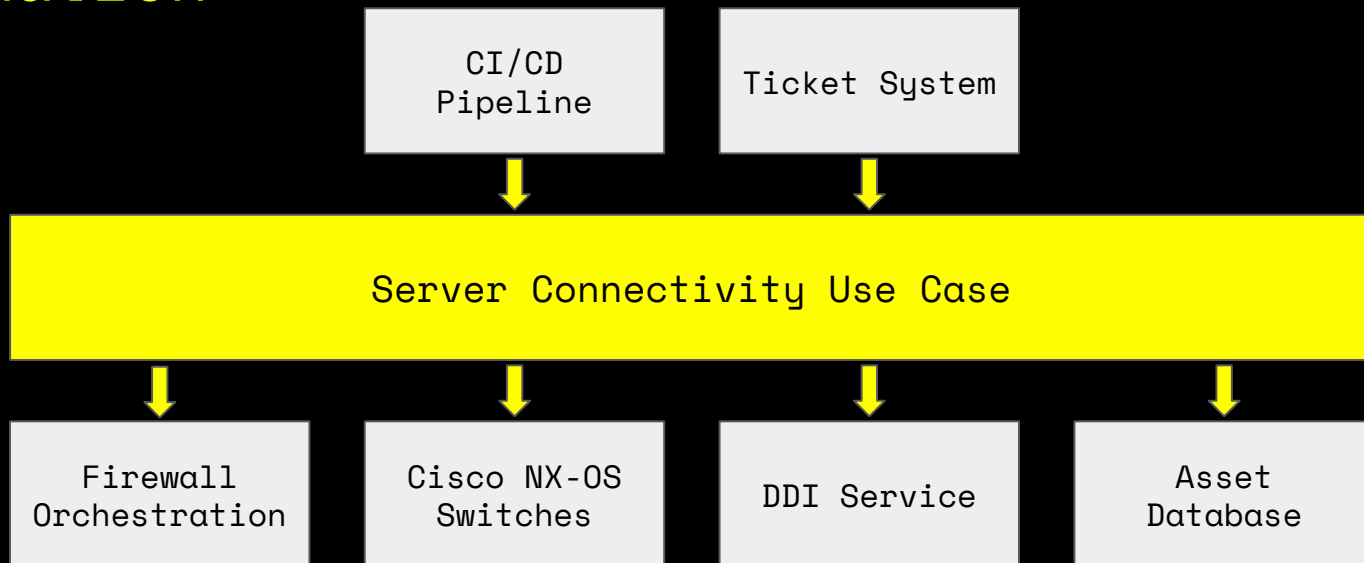
First use case: server connectivity automation



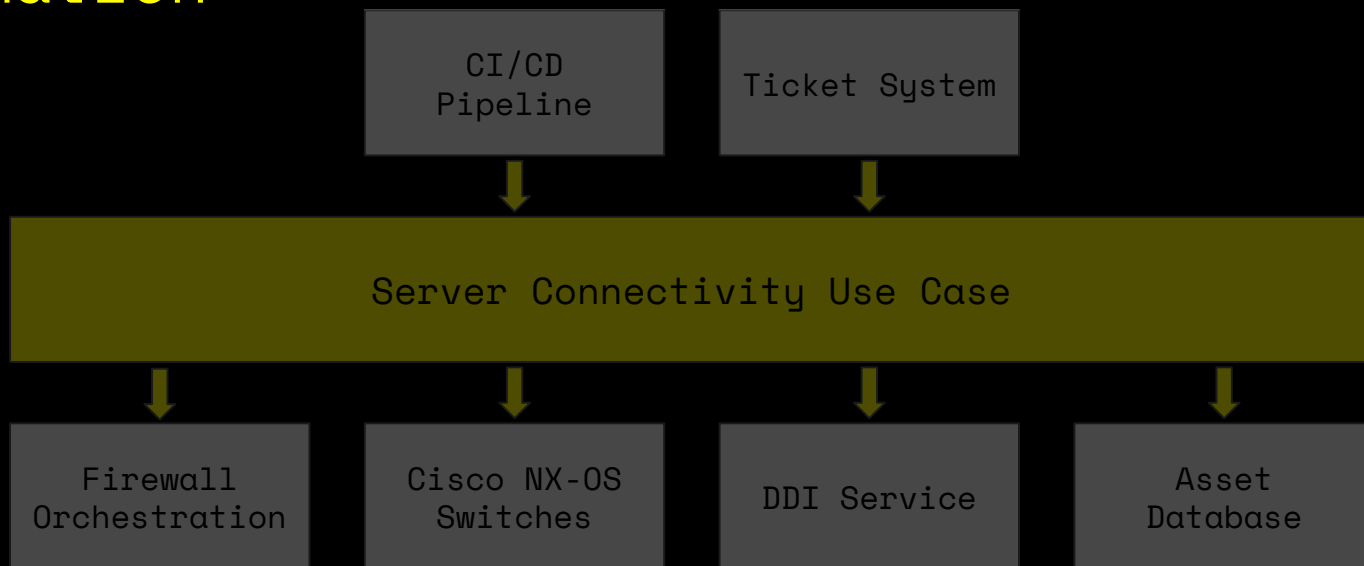
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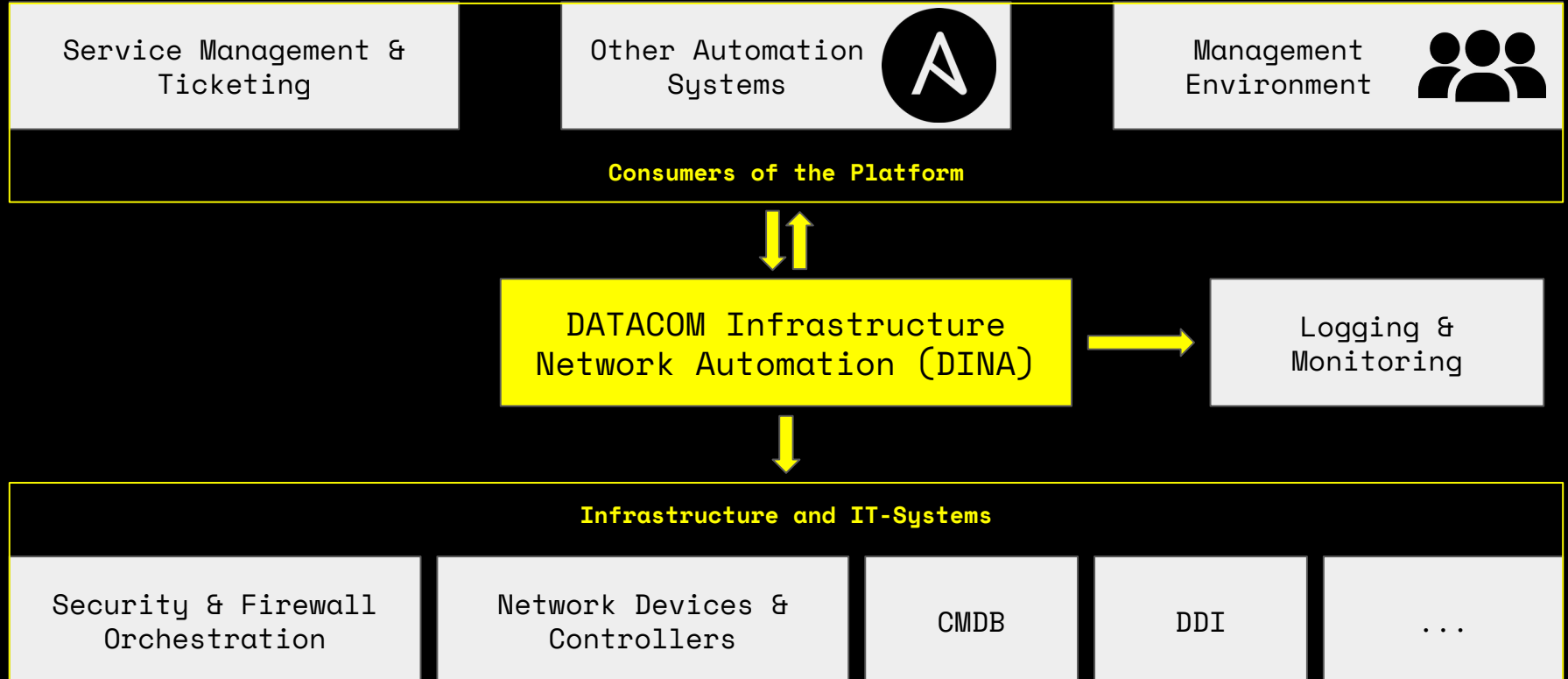
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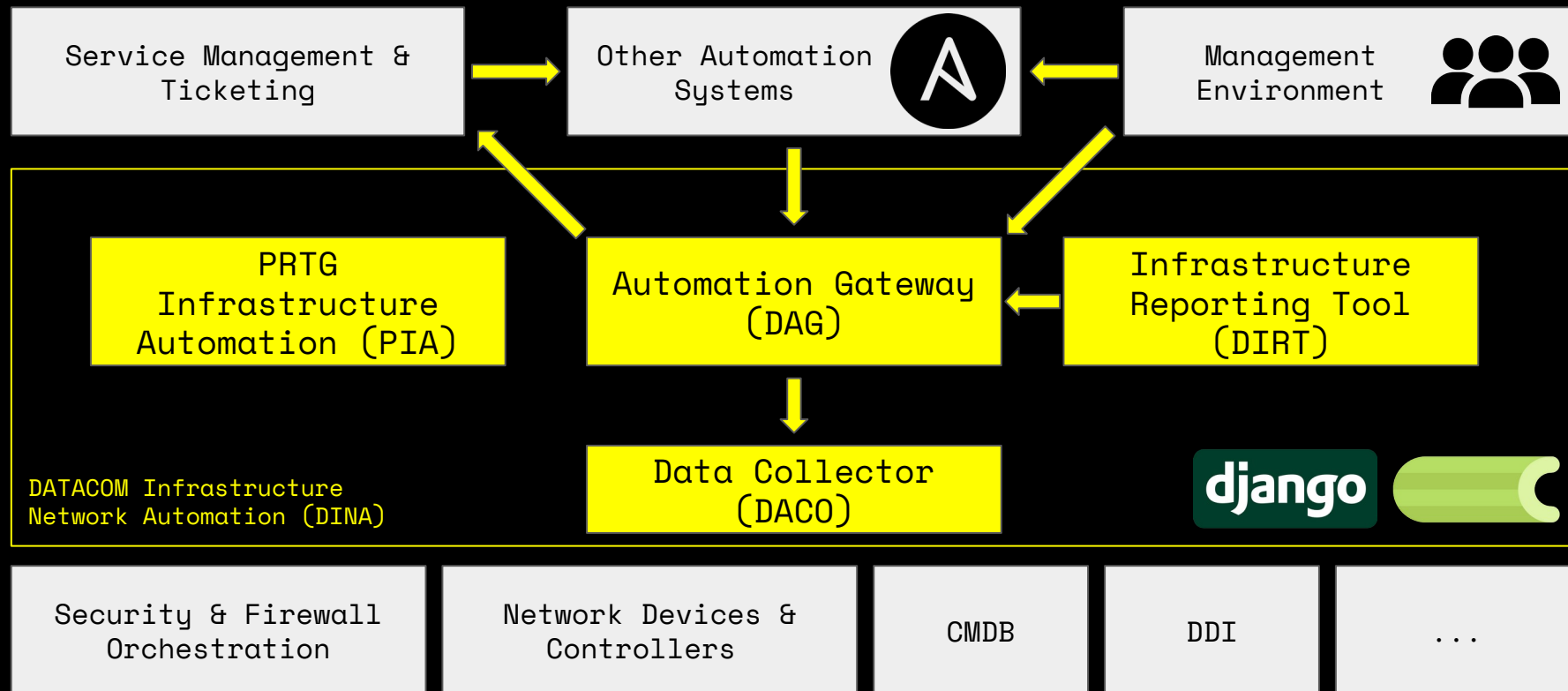
We need a development team!

How the platform
looks today

Platform Architecture



Platform Architecture



Platform Principles

Platform Principles

Air-gapped by design

Build once, deploy unchanged

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Build once, deploy unchanged

Operations and scalability **first**

Distribute workload execution with
Django and Celery

Platform Principles

Air-gapped by design

Operations and scalability **first**

Usability drives adoption

Develop without live-system access

Build once, deploy unchanged

Distribute workload execution with Django and Celery

Design every workflow from the end-user perspective

Enable fast iteration during development

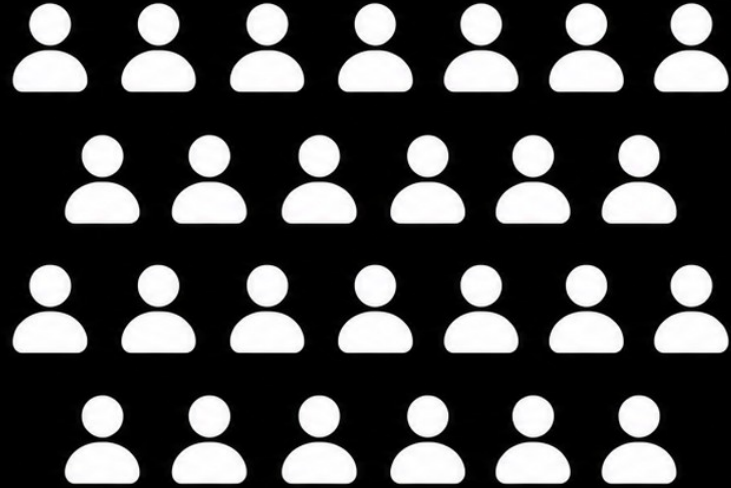
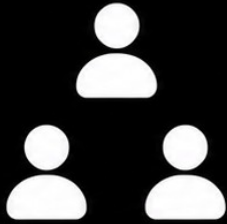
Swagger UI

api

POST	/api/central-firewall/config_interface_to_firewall/	 
GET	/api/cmdb/network-zones/	 
POST	/api/cmdb/rollout/	 
POST	/api/dns/create_host/	 
DELETE	/api/dns/record/	 
GET	/api/dns/records/	 
POST	/api/dns/reserve_ip_network/	 
PATCH	/api/dns/update_host/	 
POST	/api/mpls/tenant/bulk/rollout/	 
POST	/api/mpls/tenant/bulk/rollout_only_network/	  
POST	/api/mpls/tenant/rollout/	 
POST	/api/mpls/tenant/rollout_only_network/	 
POST	/api/network/create_in_infrastructure/	 

Scaling-out: from use
case to workflows

Revised view: Who is really the end user?



Other automation
systems and Ansible
playbooks

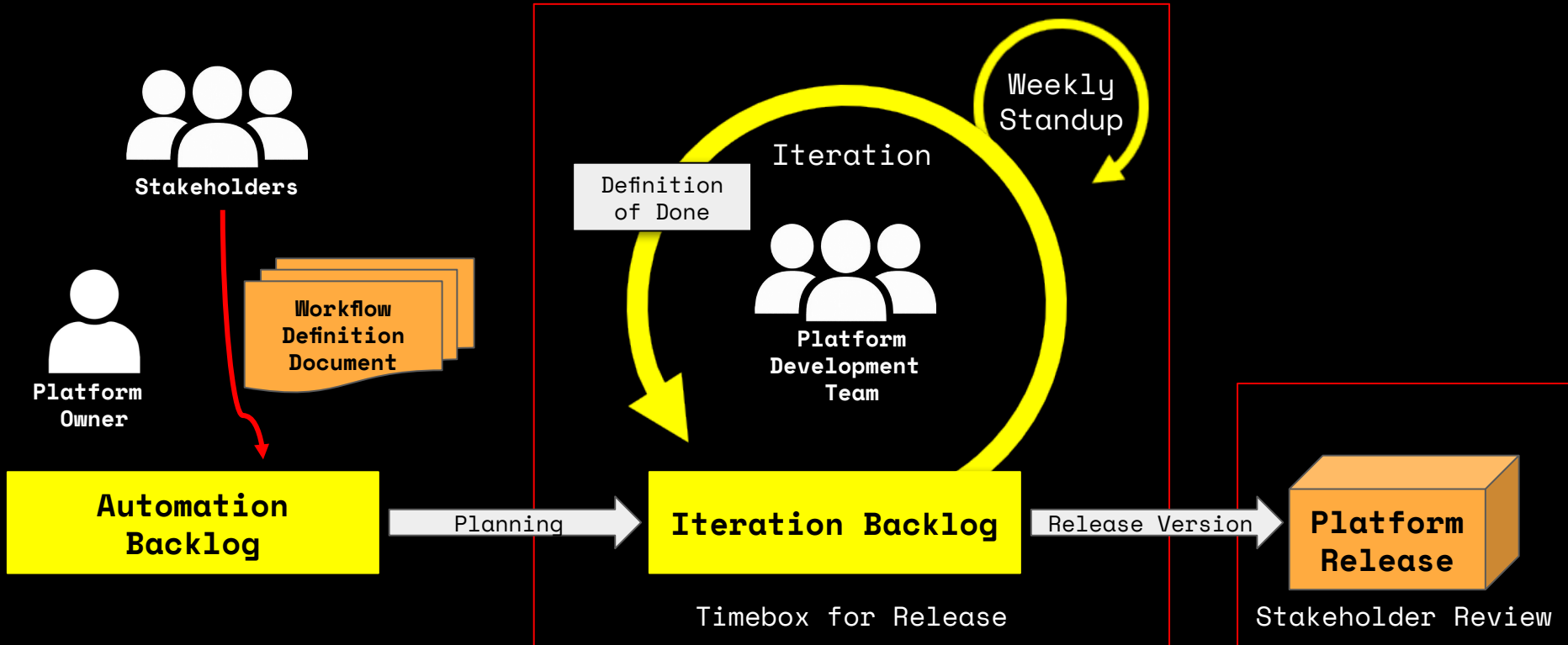
Network
Engineers/Operators
and Support Staff

Data Center Self-Service Portal

DC Switches UI			
DC Switches			
10 – 21 of 113 port groups			
Hostname	Peer-Switch	Actions	
nx-0c11-kv0f1	nx-0c11-kv0v2	List Ports	Locator ID ☐
sw-3c22-aa4e7	sw-3c22-aa4fd	List Ports	Locator ID ☐
lnx-6b12-re2f1	lnx-6b12-re2d2	List Ports	Locator ID ☐
zsw-9e11-wp9f2	zsw-9e11-wp9f1	List Ports	Locator ID ☐
sp-0d22-ku1k8	sp-0d22-ku1k4	List Ports	Locator ID ☐
ufw-4a14-yz4m4	ufw-4a14-yz4n2	List Ports	Locator ID ☐
vxd-2b9-qy7m1	None	List Ports	Locator ID ☐
flx-cd36-kv1f2	flx-cd36-kv1f1	List Ports	Locator ID ☐
s44-dd22-taa0x2	None	List Ports	Locator ID ☐
zaw-ed11-taa0x9	zaw-ed11-taa1c	List Ports	Locator ID ☐

From Use Case to Workflow Contracts

From Use Case to Workflow Contracts



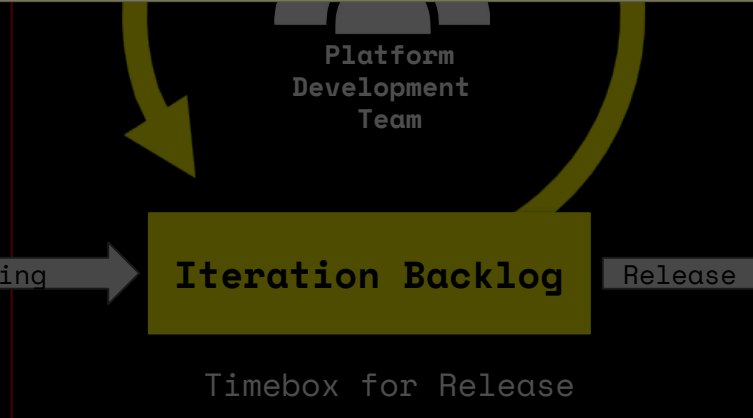
From Use Case to Workflow Contracts



Automation is continuous organizational effort



Workflow
Definition
Document



Automation
Backlog

Planning

Iteration Backlog

Release Version

Platform
Release

Stakeholder Review

From Use Case to Workflow Contracts

- One shared **Workflow Definition Document (WDD)** per use case
- Contents:
 - Goal and KPI
 - Workflow Definition (BPMN)
 - User-Interface Design
 - Data Models and Interfaces
 - Constraints
 - External Dependencies
 - ...

Cisco Vulnerability Detection Workflow Definition Document

This is an example **Workflow Definition Document (WDD)** that describes the overall goal, workflow and sub-workflows, data model, user experience, constraints, limitations and external dependencies.

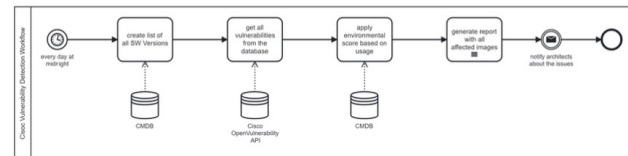
Author	Henry Ölsner
Date	2026-02-05
Status	Draft
Version	0.1
Approval	Pending
Approved by	Requester, Workflow Owner, Product Owner, Development Lead

Automation Goal

The goal of this workflow is to automate the process of detecting vulnerabilities in Cisco devices, thereby improving security and reducing manual effort for IT teams. The workflow will collect and aggregate data from various sources, analyze it for potential vulnerabilities, and creates a report that is sent by email to the IT team for further action.

Workflow Definition

The following diagram describes the workflow:



User Experience

The process should be triggered periodically (e.g., weekly) or on-demand by the IT team. The workflow will run in the background and will not require any user interaction during its execution. Once the workflow is complete, the IT team will receive an email with the vulnerability report, which they can review and take necessary actions.

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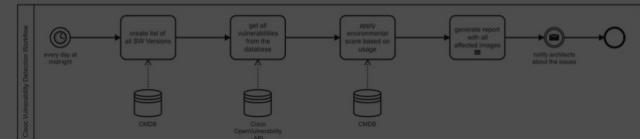
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Goal: Create a shared understanding on **what** to build

Key Takeaways



Key Takeaways

Requirements engineering and workflow contracts align users, operations, and audit needs

Automation is continuous effort that requires control, operability, and governance, not tools (in the first place)

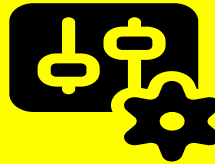
Workflow automation must shape the IT operating model and system landscape

Agile delivery balances reviewability, compliance, and urgent operational demand.

What's next?



New Workflows



**Operational
Workbench**



**On-Premises Coding
Assistance**



AI Model Hosting



Questions?

Thank You